GE Hydro

INSPECTION AND TEST PLAN

Section 1700

TURBINE GUIDE BEARING

Project: SEVEN MILE

Unit 4

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1700 : Turbine guide bearing	Page: 1701

CODE **A: Hold point/ customer** **T: Witness point/ customer**	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
1701	100	I,R	Incoming inspection.	See ITP Section 100		QCR 101	
1702	2200	O,M,R	Clean the inner oil sump and inspect contact surfaces. Prepare the O'ring. Align the intake centreline and raise inner oil sump until it slides over the outer sump studs. Install tab washers and nuts. Tight. Center the inner oil sump to the outer oil sump. Torque. Torque monitoring.	Installation proc. SMI-U04-02058 Steps 2.1 to 2.4 Drawing 222038221	Torque value is 40 N.m lub.	QCR 004 ✓	Jan 17th 2003
1703	2200	O,I,R	Cap bosses under the sump. Fill the sump with varsol to its normal level of operation. Keep the varsol in the sump for a minimum of 12 hours. Check for leaks then drain.	Installation proc. SMI-U04-02058 Step 2.5 Drawing 222038221		QCR 004 ✓	Jan 11th 2003

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 10th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1700 : Turbine guide bearing	Page: 1702

CODE **A: Hold point/ customer** **T: Witness point/ customer**	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
							—
1704	2200	O	Clean the surfaces and threaded holes. Rig and lower the sump in the pit.	Installation proc. SMI-U04-02058 Steps 2.6 to 2.8 Drawing 222038221			[signature] Jan 12th 2003
1705	2200	O	Place the guide bearing support on parallel blocks on service bay. Clean. Install adjusting bolts to bearing support plate. Rig and lower the guide bearing support into the pit on wood blocking. Cover the sump assembly and protect until after unit alignment.	Installation proc. SMI-U04-02058 Steps 2.9 to 2.12 Drawing 222038221			[signature] Jan 25th 2003.
1706	2200	I,R	Perform a pressure test on the cooling coil ~~on the erection bay.~~ Installed	Installation proc. SMI-U04-02058 Step 3.1	Pression: ~~700 KpA~~ Done at 1 MPa Duration:	QCR 005 ✓	[signature] Jan 30th 2003

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 10th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1700 : Turbine guide bearing	Page: 1703

CODE A: Hold point/ customer T: Witness point/ customer	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
				Drawing 222038322	10 min.		
1707	2200	O	Clean and install cooling coil assembly in the oil sump.	Installation proc. SMI-U04-02058 Step 3.2 Drawing 222038291			Jan 16th 2003
1708	2200	I,R	Fill the bottom of the sump with oil or varsol. Keep the oil or varsol in the sump for a period of 12 hours. Check for leaks then drain.	Installation proc. SMI-U04-02058 Step 3.2 Drawing 222038291		QCR 004	Jan 23rd 2003
1709	2200	O	Center the oil sump assembly to the shaft. Bolt temporary.	Installation proc. SMI-U04-02058 Step 3.3	After unit alignment		Jan 9th 2003

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 10th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1700 : Turbine guide bearing	Page: 1704

CODE
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Copy to customer

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Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
				Drawing 222038221			/
1710	2200	I,R	Dimensional inspection: Concentricity of oil sump with turbine shaft	Installation proc. SMI-U04-02058 Step 3.3 Drawing 222038221		QCR 2201	Jan 9th 2003
1711	2200	O	Drill and install spring pins. Install piping for the water in oil detector. Clean the sump assembly.	Installation proc. SMI-U04-02058 Steps 3.4 to 3.7 Drawings 222038221 222038291			Feb 12th 2003.
1712	2200	O	Remove the four temporary bolts. Remove the wood blocking and sit the bearing support on the head cover.	Installation proc. SMI-U04-02058			Jan 22nd 2003

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 10th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1700 : Turbine guide bearing	Page: 1705

CODE
A: Hold point/ customer
T: Witness point/ customer

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Copy to customer

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O: Assembly Operation
V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
			Center in the centering blocks.	Step 4.1 Drawing 222038221			/
1713	2200	I,R	Dimensional inspection: -Gap between centering block and bearing support	Installation proc. SMI-U04-02058 Step 4.1 Drawing 222038221		QCR 2204	Jan 22nd 2003
1714	2200	O,M,R	Remove, clean and re-install the studs and nuts. Torque. Torque monitoring.	Installation proc. SMI-U04-02058 Step 4.2 Drawing 222038221	Torque value is 1520 N.m lub.	QCR 004 ✓	Feb 5th 2003
1715	2200	O	Install the two dowels for the guide bearing support.	Installation proc. SMI-U04-02058			/

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 10th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1700 : Turbine guide bearing	Page: 1706

CODE **A: Hold point/ customer** **T: Witness point/ customer**	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
				Step 4.3 Drawing 222038221			Jan 28th 2003
1716	2200	O,M,R	Clean the guide bearing segments and the shaft journal. Install parts (P/B 210, line 7 and 8) to the bearing segments. Torque socket head capscrews. Torque monitoring.	Installation proc. SMI-U04-02058 Steps 5.1 and 5.2 Drawing 222038221	Torque value is 40 N.m lub.	QCR 004 ✓	Jan 30th 2003
1717	2200	O	Install the oil return tubes.	Installation proc. SMI-U04-02058 Step 5.2 Drawing 222038221			Feb 7th 2003.
1718	2200	O	Install and adjust the guide bearing segments.	Installation proc. SMI-U04-02058			

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 10th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1700 : Turbine guide bearing	Page: 1707

CODE **A: Hold point/ customer** **T: Witness point/ customer**	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
				Steps 5.3 and 5.4 Drawing 222038221			Feb 3rd 2003
1719	2200	I,R	Dimensional inspection: -Gap between turbine guide bearing segments	Installation proc. SMI-U04-02058 Step 5.4 Drawing 222038221		QCR 2207	Jan 30th 2003
1720	2200	O	Remove the runner blocking.	Installation proc. SMI-U04-02058 Step 5.5			Jan 29th 2003
1721	2200	O,I,R	Dimensional inspection:	Installation proc. SMI-U04-02058	At 0° and 180°		

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 10th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
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CODE A: Hold point/ customer T: Witness point/ customer	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
			Turbine guide bearing gap	Step 5.6 Drawing 222038221		QCR 2202 ✓ Done at 0° 180°	Jan 30th 2003 Feb 5th 2003
1722	2200	O,M,R	Install the bearing seals. Install the set screws. Torque monitoring	Installation proc. SMI-U04-02058 Steps 5.7 and 5.8 Drawing 222038221	Torque value is 15 N.m lub.	QCR 004	Feb 12th 2003.
1723	2200	O	Clean the cover plate and cover plate support. Do a final cleaning around the guide bearing and apply bearing oil on all parts around to prevent corrosion.	Installation proc. SMI-U04-02058 Steps 6.1 to 6.3 Drawings 222038221			Jan 29th 2003

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 10th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1700 : Turbine guide bearing	Page: 1709

CODE A: Hold point/ customer T: Witness point/ customer	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
				222038224			
1724	2200	O,M,R	Prepare and install the O'ring. Install the cover plate segments. Orientate and centre the cover plate. Bolt the cover plate to the guide bearing support. Torque. Torque monitoring.	Installation proc. SMI-U04-02058 Step 6.4 Drawing 222038221	Torque value is 40 N.m lub.	QCR 004 ✓	Feb 12th 2003.
1725	2200	I,R	Check concentricity between cover plate and the shaft in 8 points.	Installation proc. SMI-U04-02058 Step 6.5 Drawing 222038221	Tolerance is 0.50 mm	QCR 004 ✓	Feb 12th 2003.
1726	2200	O	Install dowels between cover and cover plate support.	Installation proc. SMI-U04-02058 Step 6.6 Drawing			Feb 12th 2003.

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 10th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro	INSPECTION AND TEST PLAN	
Project SEVEN MILE	Customer: B.C. Hydro	Contract: SM-70
Unit 4	Section 1700 : Turbine guide bearing	Page: 1710
CODE A: Hold point/ customer T: Witness point/ customer	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
				222038221			
1727	2200	O	Install packing ring segments, the retaining segments and hold down with capscrews. Tight the capsrews to cause a light contact between the packing segments and the shaft.	Installation proc. SMI-U04-02058 Step 6.7 Drawing 222038224			Feb 12 2003.
1728	2200	I,R	Check that feeler gauge from 0.25 to 0.50 mm goes between the packing segments and the shaft.	Installation proc. SMI-U04-02058 Step 6.7 Drawing 222038224		QCR 004 v	Feb 12th 2003.
1729	2200	O	Install the inspection covers. Install instrumentation. Install the screw plugs. Install the shaft vibration detector.	Installation proc. SMI-U04-02058 Steps 6.8, 7.1 to 7.3 Drawings 222038221			Feb 20th 2003-

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 10th, 2003	Accepted by: Date:	Revision no: 00

GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE	**Customer: B.C. Hydro**	**Contract: SM-70**
Unit 4	**Section 1700 : Turbine guide bearing**	**Page: 1711**

CODE **A: Hold point/ customer** **T: Witness point/ customer**	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
				222038223 222038224			
1730	2200	I,R	Final inspection : Ensure that all the work and documentation have been completed.	Installation proc. SMI-U04-02058		QCR 003	BB 03/24/03

RECEIVED 03/24/03

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: January 10th, 2003	Accepted by: Date:	Revision no: 00

	GENERAL INSPECTION REPORT		QCR 004
	PROJECT: Seven Mile	Unit: 4	ITP step no.: 1702

EQUIPMENT : **Torque of the nuts holding together the inner and the outer oil sumps**

The nuts installed on the studs holding together the inner and the outer oil sumps were torqued at a value of 40 N . m lubricated.

RECEIVED
JAN 1 4 2003

Measured by.:	F. Boulet	(/ /)
GE Repres.:		(Jan/17th/2003)
Cust. Repres.:		(/ /)

GE Canada International Inc.	Reference NC no.: ______ Comments: ______	Rev.: 0 Date: 07/23/01

GE	GENERAL INSPECTION REPORT	QCR 004
	PROJECT: Seven Mile Unit: 4	ITP step no.: 1703

EQUIPMENT : **Leak test in the turbine guide bearing sump**

The turbine guide bearing has been filled with varsol and rested like this during 12 hours.
No leak has been observed.

Measured by.: F. Boulet (/ /)
GE Repres.: Jan 11th/2003
Cust. Repres.: (/ /)

GE Canada International Inc.	Reference NC no.: Comments:	Rev.: 0 Date: 07/23/01

	PRESSURE TEST	QCR 005
	PROJECT: Seven Mile Unit: 4	ITP step no.: N/A

Around 1706

VERIFIED EQUIPMENT(S): **Turbine guide bearing cooling coils**

TEST DESCRIPTION : Test done with water after installation in the sumps

PRESSURE : **1 Mpa** **DURATION :** **1 hour**

ACCEPTED TEST	**X**
REJECTED TEST	

COMMENTS :

RECEIVED JAN 3 0 2003

Measured by.:	P. Lamontagne	(Jan/25th/2003)
GE Repres.:	[signature]	(Jan/29th/2003)
Cust. Repres.:		(/ /)

GE Canada International Inc.	Reference NC no.: Comments:	Rev.: 0 Date: 23/07/02

	GENERAL INSPECTION REPORT	QCR 004
	PROJECT: Seven Mile Unit: 4	ITP step no.: 1708

EQUIPMENT : **Leak test in the turbine guide bearing sump - in pit**

After the installation of the turbine guide bearing, it has been filled with varsol and rested like this during 12 hours. No leak has been observed.

JAN 23 2003

Measured by.:	P. Lamontagne	Jan/20th/2003
GE Repres.:	[signature] P. Lamontagne	(Jan/23rd/2003)
Cust. Repres.:		(/ /)

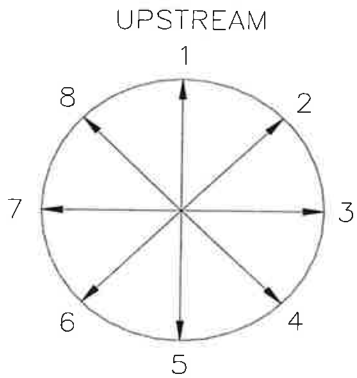
GE Canada International Inc.	Reference NC no.: Comments:	Rev.: 0 Date: 07/23/01

CONCENTRICITY OF INNER OIL SUMP WITH TURBINE SHAFT

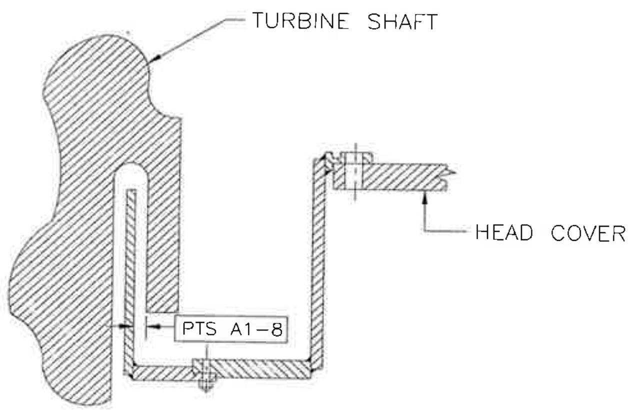
QCR 2201

PROJECT: Seven Mile | Unit: 4 | ITP step no.: 1710

Measurement Instruments	Ambient Temperature
Snap gauges	Before: 65°F
Caliper 417053	After:



PTS	A (mm)
1	35.77
2	35.66
3	35.70
4	35.48
5	35.46
6	35.82
7	35.75
8	35.78



	TOLERANCE (mm)	CALCULATED (mm)
Concentricity	0.5	0.14

Measured by.:	Rob Speers	(Jan/9th/2003)
GE Repres.:	Don Burrell [illegible]	(Jan/9th/2003)
Cust. Repres.:		(/ /)

GE Canada International Inc.

Reference NC no.:

Comments:

Rev.: 0

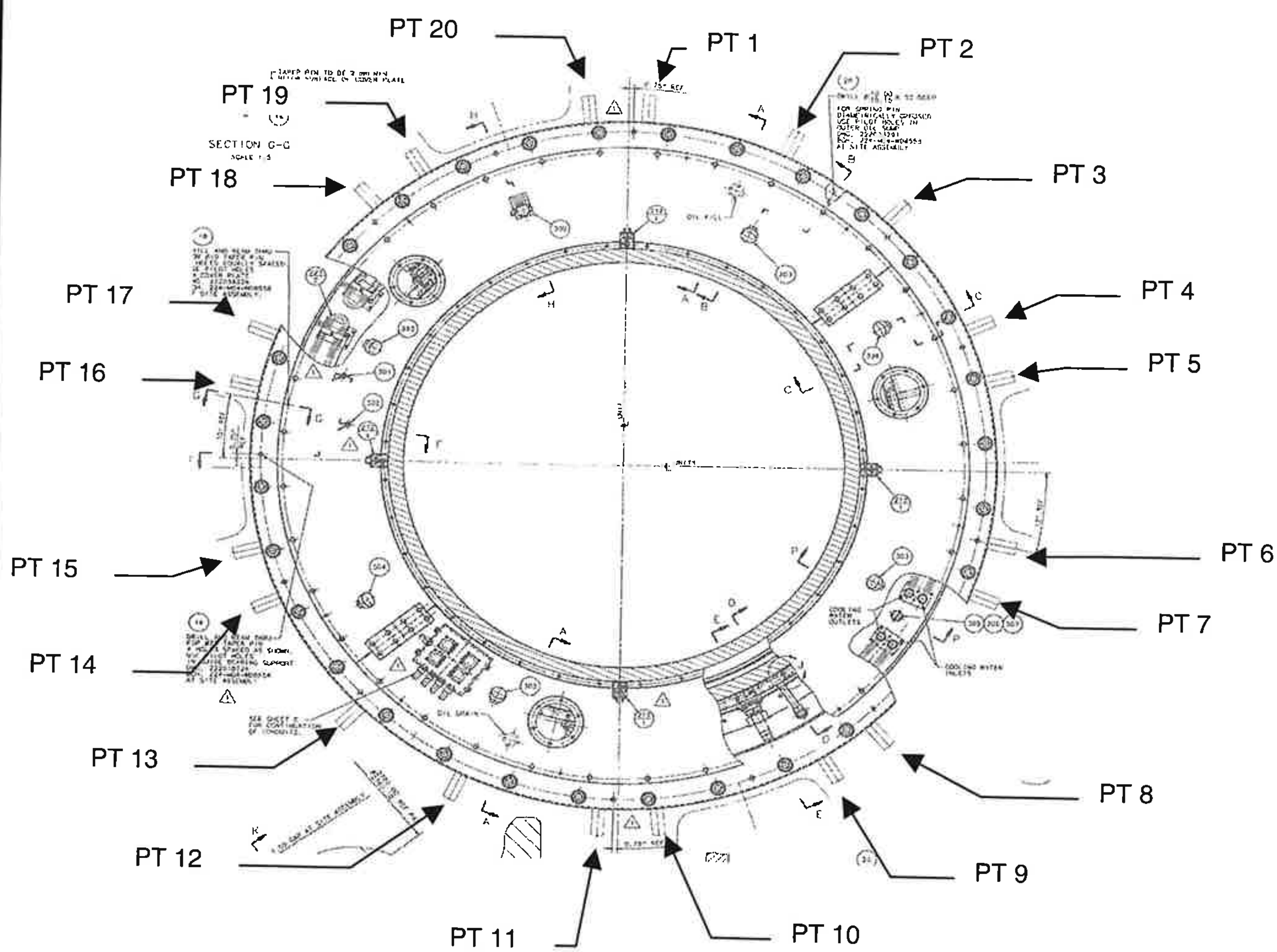
Date: 02-12-09

GE	GAP BETWEEN CENTERING BLOCKS AND GUIDE BEARING FLANGE	QCR 2204
	PROJECT: Seven Mile Unit: 4	ITP step no.: 17/3

Measurement Instruments	Ambient Temperature
Feeler gauge	Before:
	After:

	TOLERANCE (mm)	CALCULATED (mm)
Max variation to average	±0.05	

NOMINAL RADIAL GAP : 0.08 to 0.25mm



RECEIVED JAN 22 2003
Clt

Measured by.:	Marc Kitchin	(Jan/21st/2003)
GE Repres.:	[signature]	(Jan/21st/2003)
Cust. Repres.:		(/ /)

GE Canada International Inc.	Reference NC no.: Comments:	Rev.: 1 Date: 03-01-21

GAP BETWEEN CENTERING BLOCKS AND GUIDE BEARING FLANGE	QCR 2204
PROJECT: Seven Mile Unit: 4	ITP step no.: 17/3

Measurement Instruments	Ambient Temperature
Feeler gauge.	Before:
	After:

PTS	A * (mm)	Variation to average (A - average) (mm)
1	0,038	0
2	0,076	0,036
3	<0,038	0
4	<0,038	0
5	<0,038	0
6	0,051	0,011
7	<0,038	0
8	<0,038	0
9	<0,038	0
10	<0,038	0
11	<0,038	0
12	<0,038	0
13	<0,038	0
14	<0,038	0
15	<0,038	0
16	<0,038	0
17	<0,038	0
18	<0,038	0
19	<0,038	0
20	<0,038	0

Average: 0,04

* The average has been calculated using 0,038 for the values <0,038.

RECEIVED JAN 22 2003

CH

Measured by.:	Marc Kitchin	(Jan/21st/2003)
GE Repres.:	[illegible]	(Jan/21st/2003)
Cust. Repres.:		(/ /)

GE Canada International Inc.	Reference NC no.: Comments:	Rev.: 1 Date: 03-01-21

	GENERAL INSPECTION REPORT	QCR 004
	PROJECT: Seven Mile Unit: 4	ITP step no.: 17|4

EQUIPMENT : **Torquing of the turbine guide bearing sump on headcover**

The turbine guide bearing sump was torqued on the headcover at a value of 1520 N.m lubricated.

Measured by.: John Willman (Jan/28th/2003)

GE Repres.: (Feb/3rd/2003)

Cust. Repres.: (/ /)

GE Canada International Inc.	Reference NC no.: ____ Comments: ____	Rev.: 0 Date: 07/23/01

	GENERAL INSPECTION REPORT	QCR 004
	PROJECT: Seven Mile Unit: 4	ITP step no.: 1716

EQUIPMENT : **Torquing of the retaining plates on the TGB segments**

The capscrews (bill 210, line 27, 28) holding the retaining plate (line 7) on the turbine guide bearing segments were torqued at a value of 40 N.m lubricated.

RECEIVED
JAN 30 2003

Measured by.:	P. Lamontagne	(Jan 25th 2003)
GE Repres.:		(Jan 29th 2003)
Cust. Repres.:		(/ /)

GE Canada International Inc.	Reference NC no.: ______ Comments: ______	Rev.: 0 Date: 07/23/01